Beyond Chatbots

Exploring AI Coding Tools for Teaching, Learning, and Research

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WORKSHOP MAP

You will move from **using** AI tools to **planning** one of your own

Beyond Chatbots	Teaching & Learning	Research	Ideas to Applications	Critical Checks & Q&A

01

Explain

How chatbots differ from AI coding tools.

02

Identify

Uses in teaching, learning and research.

03

Plan

A small tool or automation from your own idea.

04

Check

Accuracy, security, ethics and validity.

The core shift: **using** AI tools → **creating** tools with AI.

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PART

01

Beyond Chatbots

What AI coding tools are, and how they differ from the chatbots we already know.

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THE SHIFT

AI is no longer only something we **use**; it is something we can **build with**

UNTIL NOW

Using AI tools

Ask · Discuss · Generate

You type. The AI answers. You copy the result somewhere else.

THIS WORKSHOP

Creating tools with AI

Build · Test · Automate

You describe. The AI builds. You test, judge and improve.

> You do **not** need to be a programmer. You need to know your learners, your research question and what "good" looks like.

From **using** AI tools to **creating** tools with AI.

Jeon & Lee (2023).

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KEY MESSAGES

Three kinds of tool, three kinds of **activity**



AI Chatbots

Ask questions, talk ideas through and generate content.

ASK · DISCUSS · GENERATE



AI Coding Tools

Describe what you need. The tool writes, runs and fixes the code.

BUILD · TEST · AUTOMATE



Educational Applications

What you make: tools that serve your class or your study.

TEACH · LEARN · RESEARCH

Coding tools are the **means**; educational applications are the **goal**.

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HOW THEY DIFFER

From chatbot to agent to **coding agent**: each step does more on its own

STEP 1 · WEB CHATBOT

Answers

Replies in a chat window. You copy the result and use it somewhere else.

STEP 2 · AGENT

Acts

Plans steps and uses tools for you: search, files, other apps.

STEP 3 · CODING AGENT

Builds

Reads your project, writes and runs code, tests it and fixes it.

Same underlying AI; the difference is what it is **allowed to do**.

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HOW THEY DIFFER · SIDE BY SIDE

Same request, different result: text to copy versus a working file

CHATBOT

Make a vocabulary quiz for my class.

Here is a 10-question quiz and some code. Copy it into a file, save it as .html, then open it.

Coding agent

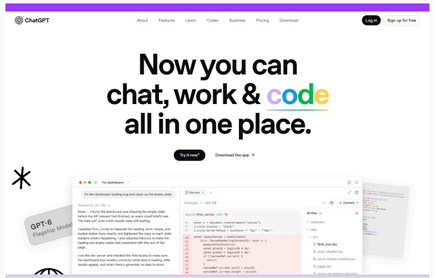
```
> Make a vocabulary quiz for my class.
Plan: word list · quiz page · scoring
✓ created quiz.html
✓ opened it and ran a test round
! fixed: score did not reset
✓ ready — open quiz.html
```

The chatbot gives you **material**. The coding agent gives you a **tool**, tested.

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THE TOOLS YOU WILL HEAR ABOUT

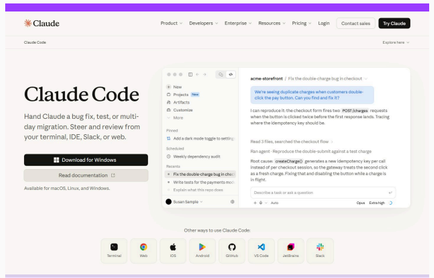
Different names, one idea: an assistant that can act in your project



ChatGPT

A chat assistant, with coding and work tools added.

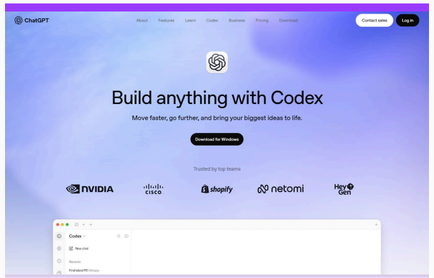
WEB · DESKTOP · MOBILE



Claude Code

A coding agent that reads your project and runs commands.

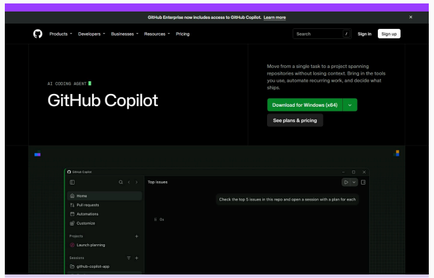
TERMINAL · IDE · DESKTOP · WEB



Codex

OpenAI's coding agent, included with ChatGPT plans.

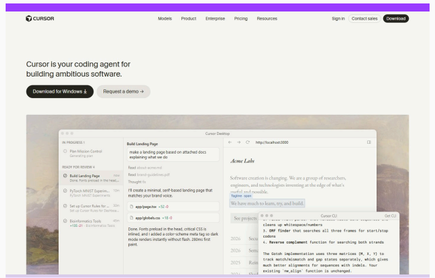
APP · IDE · CLI · CLOUD



GitHub Copilot

GitHub's coding agent, tied to issues and pull requests.

DESKTOP APP · CLI · VS CODE



Cursor

A coding agent built around an editor, with CLI and cloud agents.

DESKTOP · CLI · SLACK

Choose by the **job** — ask, discuss or build — not by the brand.

Screenshots: official product websites.

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UNDER THE HOOD

A coding agent works in a **loop** that you can supervise at every turn



YOU DECIDE

The goal, the pedagogy, the data, and what "good" means.

THE AGENT DOES

The typing, the running, the first round of debugging.

If you cannot judge the result, you are not yet ready to rely on it.

You stay in the loop; the agent does the **typing**.

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02

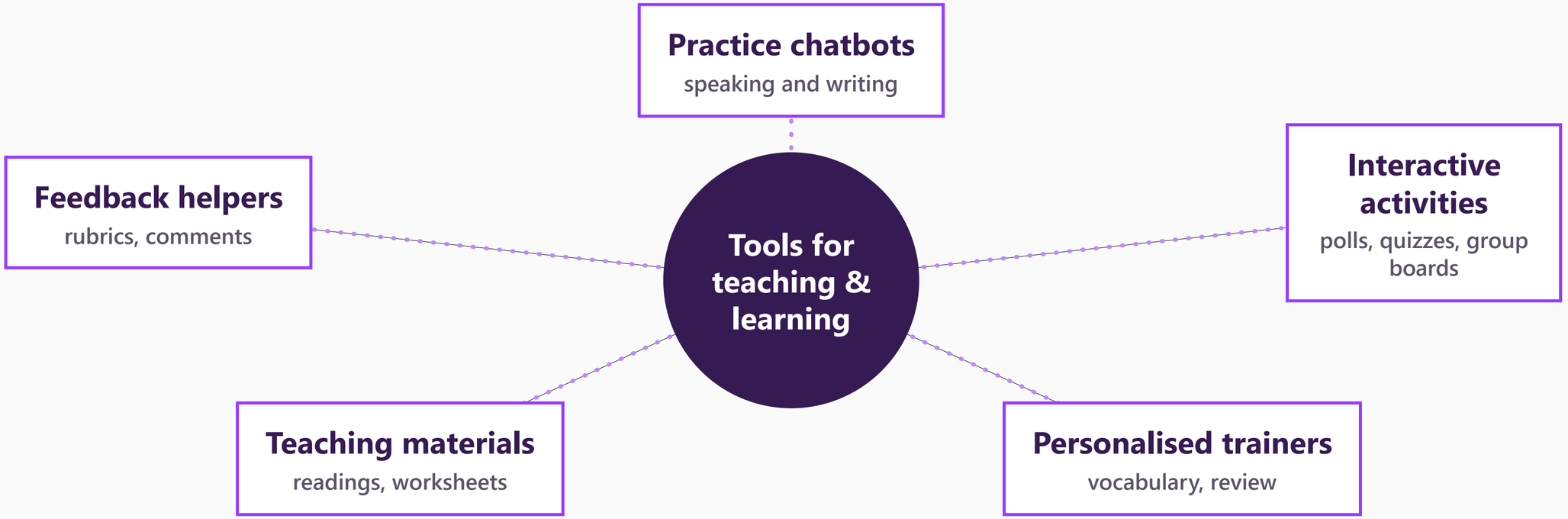
AI Tools for Teaching & Learning

Practice chatbots, interactive activities and personalised learning tools — three sample cases.

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TEACHING & LEARNING · THE MAP

One coding tool can produce **many kinds** of teaching and learning application



Start from the **learner activity** you want, then pick the shape of the tool.

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Kasneci et al. (2023).

CASE 1 · PRACTICE CHATBOT

A chatbot becomes **pedagogy** when the teacher sets its rules

Level
A2 to B2 changes sentence length and the hints.

Feedback style
Recast, explicit tip, or no correction.

Wait time
A hint appears after 5 or 10 seconds of silence.

First-language support
A short Korean gloss for beginners, on request.

ONE LEARNER MESSAGE, TWO FEEDBACK STYLES

I want coffee.

RECAST
Oh, a coffee — sure! What size would you like?

EXPLICIT
Tip: say “a coffee” or “some coffee”.

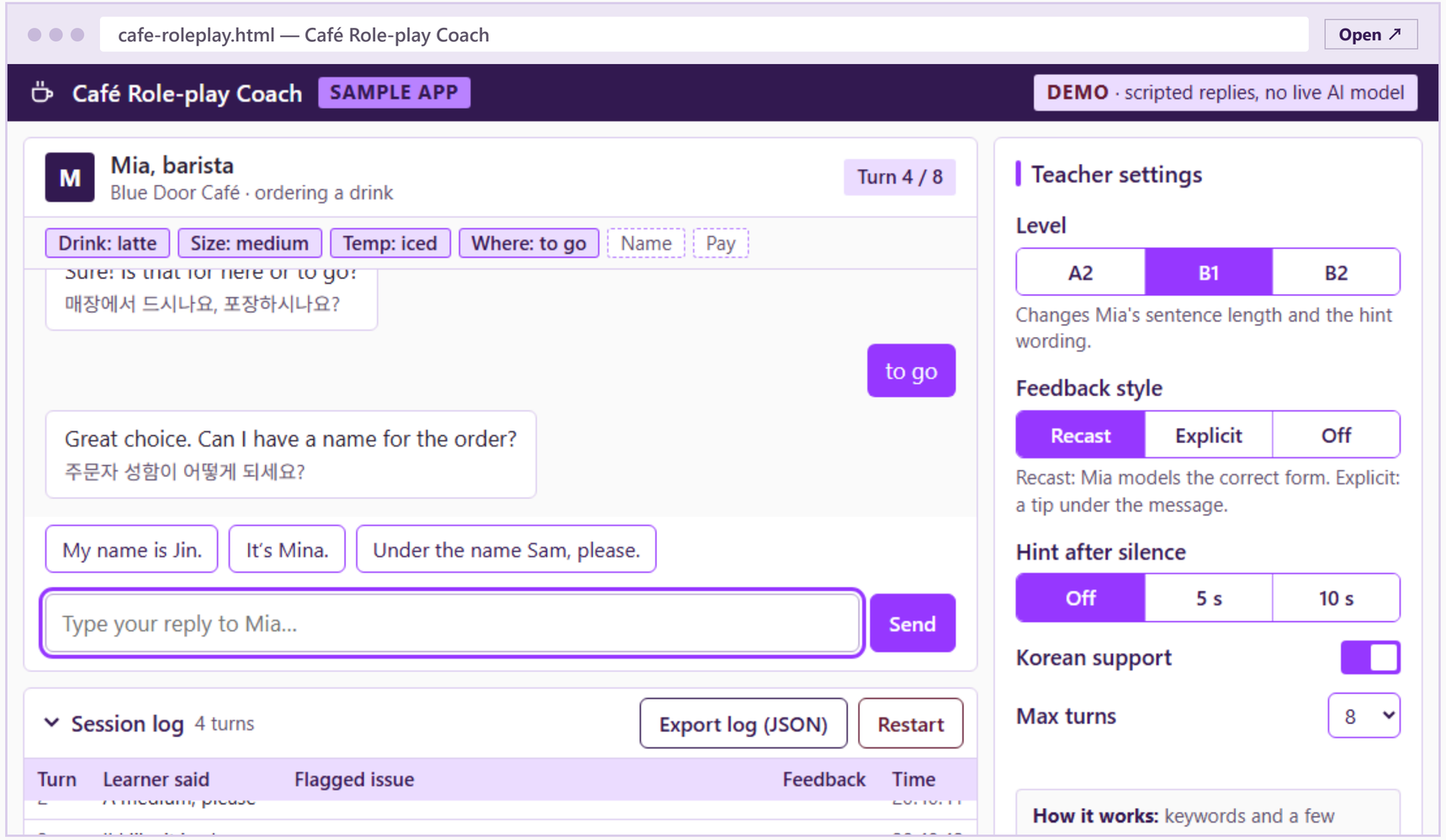
Feedback style, level and wait time are **teaching decisions**, not technical ones.

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Jeon, Lee & Choe (2023).

SAMPLE TOOL · TEACHING

Café Role-play Coach: a speaking-practice chatbot with **teacher-set rules**



TRY IT

1. Order a drink. Type **"I want coffee"**.
2. Switch feedback from **Recast** to **Explicit**.
3. Turn on **Korean support** and ask for a hint.

What to notice: the replies come from a short script and a few rules. No AI model runs here; the *design* is the point.

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CASE 2 · INTERACTIVE LEARNING ACTIVITY

A class can become **interactive** on students' own phones

Teacher's screen
The slide deck opens an activity.

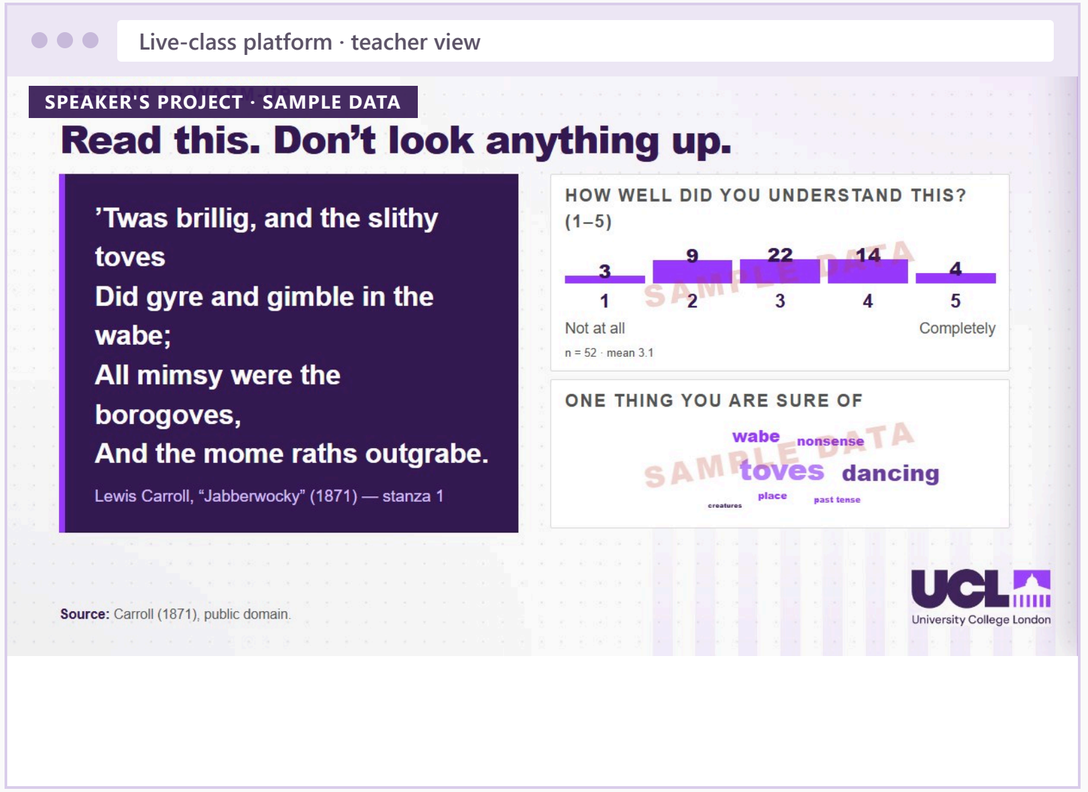
↓

Shared database
Answers arrive live, anonymously.

↓

Students' phones
Scan a QR code, then respond.

Poll | Quiz | Word cloud | Group board | Reading task



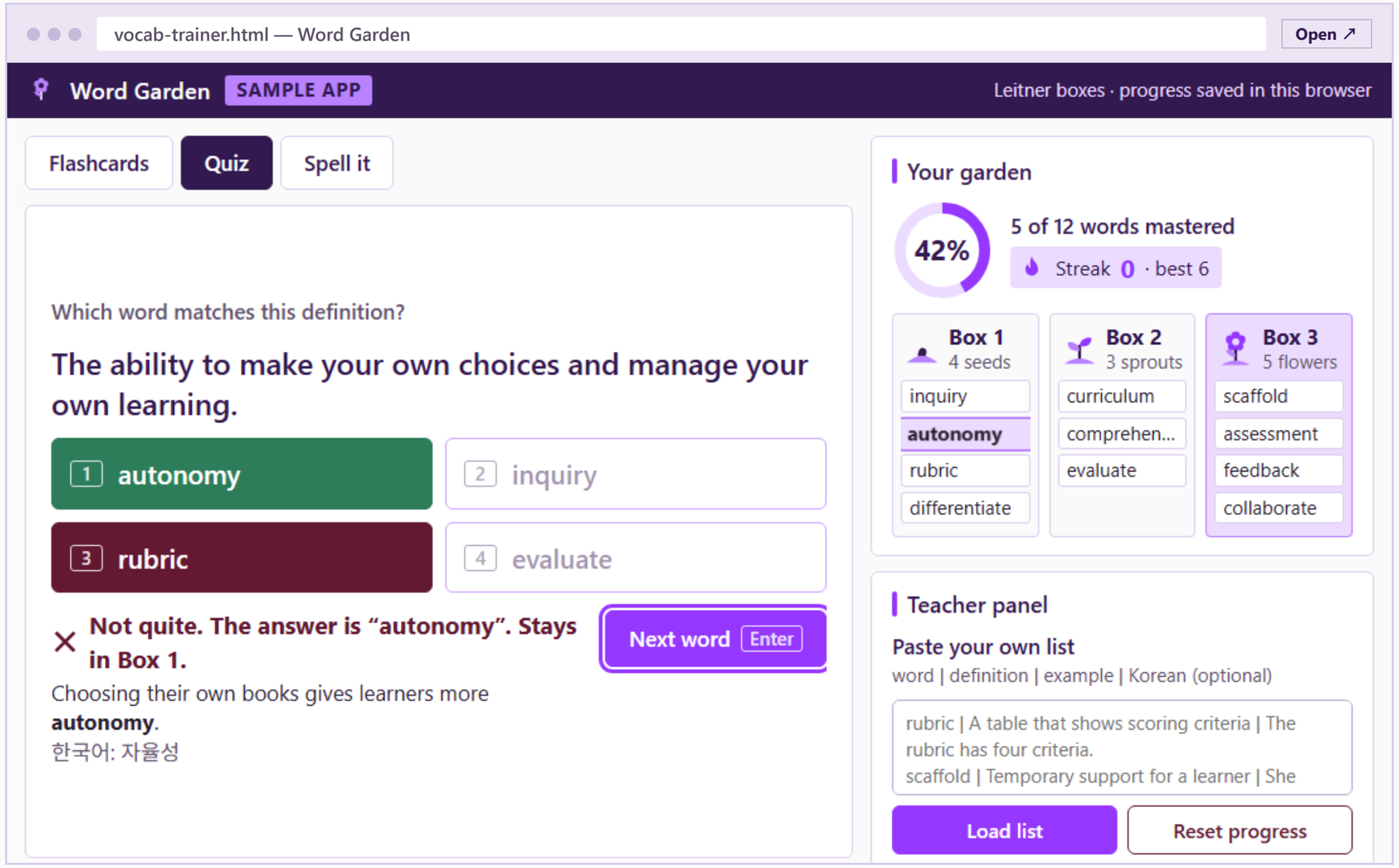
Teacher view: class responses appear on the shared screen.

Polls, quizzes and group boards in **one platform**, designed around the lesson.

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SAMPLE TOOL · LEARNING

Word Garden: a vocabulary trainer that adapts to each learner



TRY IT

1. Open **Quiz** and answer one word.
2. Get one wrong. Watch the word fall back to **Box 1**.
3. Paste your own list in the **Teacher panel**.

What to notice: three Leitner boxes decide what you see next. The teacher's word list goes in; the learner's progress stays in the browser.

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WHY BUILD?

Building your own tool puts the **pedagogy** back in your hands, at a price

	Asking a general chatbot	A tool you build
Pedagogy	Re-explained in every chat	Built into the settings
Consistency	Varies from chat to chat	Same rules for every learner
Records	Scattered chat histories	A log you design and export
Control	The provider's interface	Your layout, your rules
Effort	Minutes	Hours to days, plus testing and upkeep

Build when the activity is **repeated** and the rules **matter**.

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YOUR TURN · TEACHING & LEARNING

Start from the **learner's activity**, not from the tool

MY LEARNERS .....

COULD .....

AND THE TOOL WOULD .....

THINK · 1 MIN

Finish the three lines for one activity from your own class or study. Then tell a neighbour.

1 min

If you can finish those three lines, you have the **start of a spec**.

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03

AI Tools for Research

Instruments, data collection, analysis and the routine work around research.

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RESEARCH · THE MAP

Research tools follow the **research workflow**, from instrument to report

Instrument → Collect → Analyse → Automate → Report

Instrument

An online task or questionnaire with your own logic.

Collect

Time-stamped logs and a clean CSV export.

Analyse

Word counts, concordances, descriptive statistics.

Automate

Clean transcripts, rename and convert files in batch.

Report

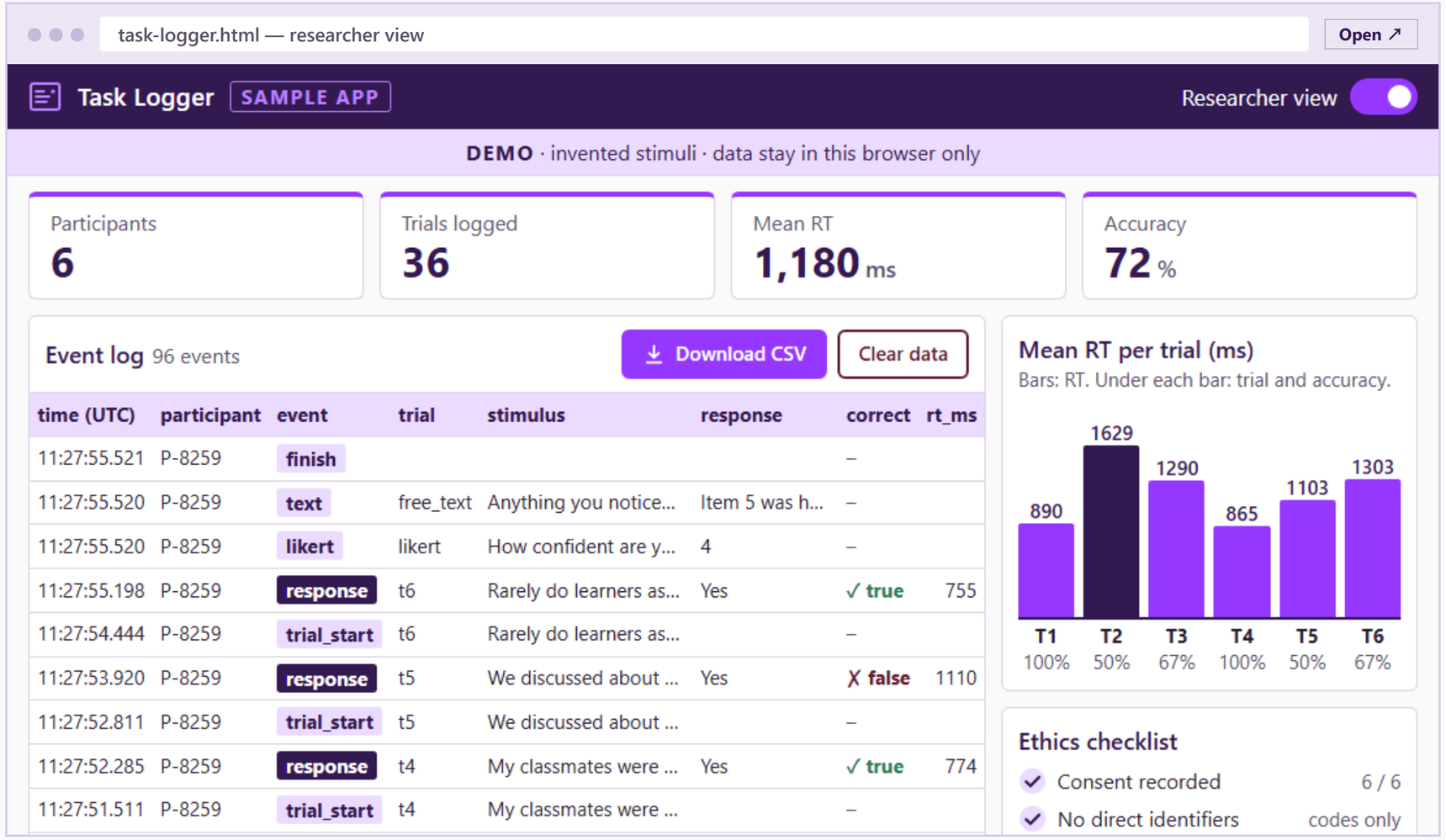
Tables and figures you can regenerate in seconds.

Automating a step is easy to **start** and important to **verify**.

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SAMPLE TOOL · RESEARCH INSTRUMENT

Task Logger: a browser task that records every event with a time stamp



TRY IT

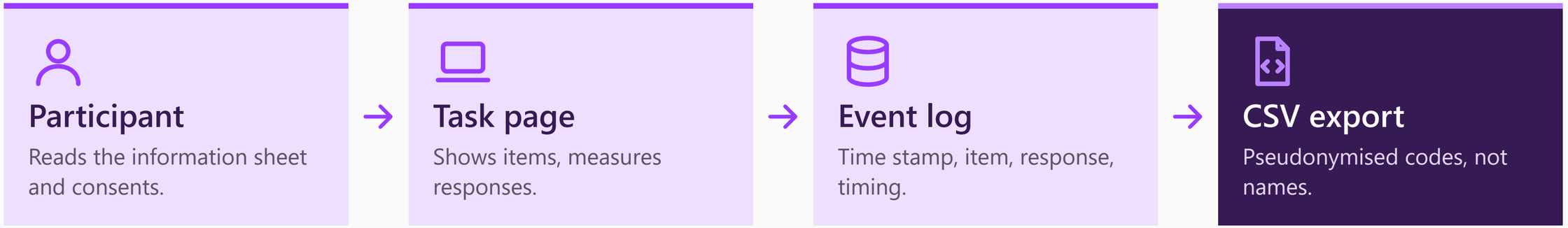
1. Read the **researcher view**: events, timings, accuracy.
2. Switch the toggle off and **run the task** as a participant.
3. Return and press **Download CSV**.

What to notice: consent comes first, no names are collected, and the export is pseudonymised. All data here are invented.

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RESEARCH INSTRUMENT · DESIGN

A data-collection tool is only as **ethical** as its design



Consent first

Nothing is recorded before the participant agrees.

No direct identifiers

Participant codes, not names or emails.

Stored where approved

Match the storage to your ethics approval.

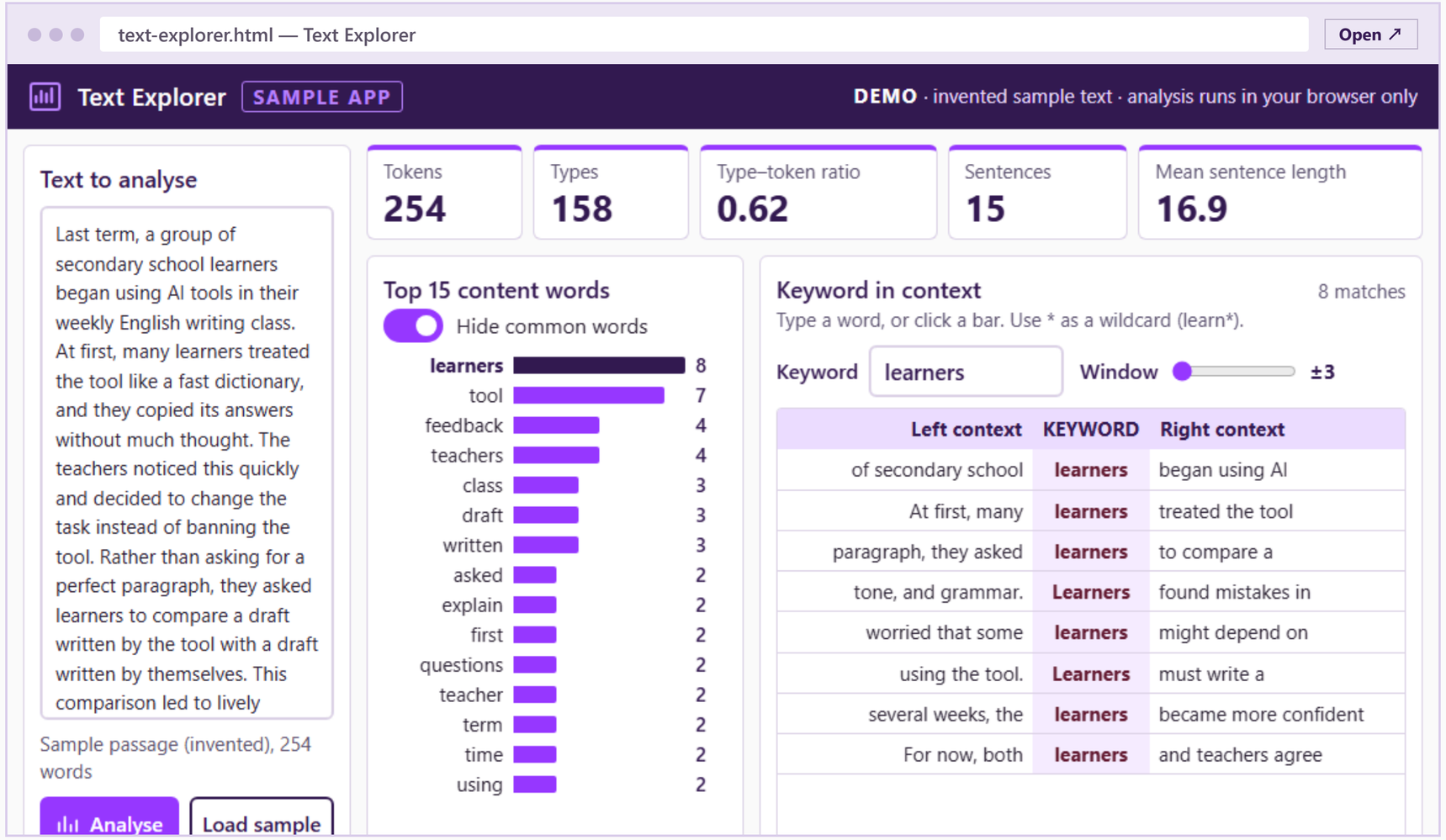
Ethics approval covers the whole tool: where data go, who sees them, how long they stay.

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BERA (2024), Ethical Guidelines for Educational Research (5th ed.).

SAMPLE TOOL · ANALYSIS

Text Explorer: word frequencies and concordance lines in your browser



TRY IT

1. Click a bar to search that word in context.
2. Turn **Hide common words** off and on.
3. Widen the **window** to see more context.

What to notice: counting is easy to automate; interpreting what the counts mean is still yours.

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RESEARCH · AUTOMATION

Repetitive research chores become a request you can review

Coding agent · data folder

```
> Clean the 120 interview transcripts: remove time
stamps, replace names with codes, save to /clean, write
a log.
Plan: read 3 samples · write script · dry run · run
✓ processed 120 files, 0 errors
✓ 14 names replaced by P01–P14 (key saved separately)
! 3 files had no speaker labels — see log.csv
```

YOU CHECK

Three files by hand, before you trust the other 117.

YOU KEEP APART

The key that links names to codes, away from the data.

YOU CAN REPEAT IT

The script and the log document exactly what was done.

The agent saves the **hours**; the checking stays **yours**.

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RESEARCH · SPEAKER'S WORKFLOW

An AI-assisted literature workflow still needs a **human gate** at every step

Search → Screen → Summarise → Verify → Draft

Search	Screen	Summarise	Verify	Draft
AI proposes	**You decide**	**AI drafts**	**You verify**	**You write**
Candidate papers for **your** question.	Which papers meet your criteria.	Summaries. You check them against the paper.	Every reference against the record, not an AI answer.	AI may edit for clarity, never invent evidence.

Check each citation against the **record** (publisher page or DOI), not the AI's answer.

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RESEARCH · RESPONSIBILITY

Delegate the routine, keep the judgement

ROUTINE · LOWER STAKES

Delegate

Renaming and converting files, formatting, first-draft code.

JUDGEMENT · LOWER STAKES

Think together

Brainstorming, wording, alternative explanations.

ROUTINE · HIGHER STAKES

Delegate, then verify

Transcript cleaning, statistics scripts, reference lists.

JUDGEMENT · HIGHER STAKES

Keep human

Research design, interpretation, ethics, the claims you publish.

The higher the stakes, the more of the work **stays with you**.

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PART

04

From Ideas to Applications

How an idea in your head becomes a tool that works — and how to ask for it.

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THE PROCESS

From idea to application in **six steps**, with you at the controls

STEP 1

Idea

One sentence

The learner activity or research question.

STEP 2

Spec

Write it down

Who, what, rules, data, "done when".

STEP 3

Prototype

AI builds

A first version in minutes.

STEP 4

Test

Try it

With sample learners or invented data.

STEP 5

Refine

One change

Change one thing at a time.

STEP 6

Share

Pilot

Document it and get approval.

> The AI is fastest at step 3. **Everything around it is yours.**

The scarce skill is not typing code; it is **knowing what to ask for** and how to check it.

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THE PROCESS · STEP 2

A good request is a short **spec**, not a clever prompt

Goal

A café role-play chatbot for speaking practice.

Users

Adult learners, A2–B1, first language Korean.

Pedagogy

Recast errors. Offer a hint after 10 seconds of silence.

Content and data

A scripted café dialogue. No real learner data.

Limits

One HTML file. Works offline. Exports the session log.

Done when

A learner can order and pay in eight turns.

If you can write the spec, the AI can write the **first version**.

THE PROCESS · STEP 3

From one request to a working tool, with your judgement in between

Coding agent · project folder

```
> Build a café role-play chatbot for A2–B1 learners. Recast
errors, hint after 10 s, Korean gloss on request. One HTML file,
works offline, exports the log.
Plan: dialogue script · error rules · settings panel · session
log
✓ wrote cafe-roleplay.html
✓ ran six test conversations
! fixed: hint timer kept running after the turn
✓ ready — open cafe-roleplay.html
```

YOU WROTE

The spec: goal, pedagogy, limits.

THE AGENT DID

Planned, wrote, ran tests, fixed one bug.

YOU DECIDE

Is it good enough for my learners?

The request is **yours**; the first draft is the agent's; the verdict is **yours** again.

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THE PROCESS · STEPS 4 AND 5

Each round of feedback adds a **pedagogical decision**

VERSION 1

v1

Replies only

It works, but the replies feel random and nothing is recorded.

VERSION 2

v2

Adds feedback

Recast and explicit tips, plus a hint after silence.

VERSION 3

v3

Adds teacher controls

Level, Korean gloss and an exportable session log.

> Change one thing, test it, keep it or undo it.

Small steps are easier to **test** than one giant request.

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YOUR TURN · FROM IDEAS TO APPLICATIONS

Try it: write a **spec card** for a tool of your own

WHO IS IT FOR? .....

WHAT DOES IT DO? .....

WHAT MUST IT NOT DO? .....

DONE WHEN... .....

WRITE · 2 MIN

Fill in the card for your idea. Swap with a neighbour and ask: what would you test first?

2 min

A spec that a neighbour can **test** is a spec the AI can build from.

PART

05

Critical Considerations & Q&A

Accuracy, security, research ethics and educational validity — then your questions.

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CRITICAL CONSIDERATIONS

Four checks before you trust what an AI has built

Accuracy

Is the content correct, and is the result right?

CHECK AGAINST A SOURCE

Security & privacy

Where do keys, data and logs go?

NO REAL LEARNER DATA IN PROMPTS

Research ethics

Is there consent, approval and transparency?

APPROVAL COVERS THE TOOL

Educational validity

Does it help learning, or only look good?

TEST WITH REAL LEARNERS

In a field experiment with high-school maths students, unrestricted GPT access raised practice scores by 48% but **lowered exam scores by 17%** once access was removed (Bastani et al., 2025).

A tool that *feels* helpful can still fail the learner.

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Kasneci et al. (2023); Bastani et al. (2025, PNAS); BERA (2024).

CRITICAL CONSIDERATIONS · HOW TO CHECK

Treat every AI-built tool as a **prototype** until it passes your own gates

GATE 1

Run it

Does it work, and does it fail safely?

→

GATE 2

Known cases

Does it give the right answers where you know them?

→

GATE 3

Peer check

Does a colleague agree it is sound?

→

GATE 4

Pilot

Does it help real learners or real data?

→

GATE 5

Document

Record what was built, how, and with which AI.

Until a tool passes your gates, call it a **prototype** — and say so to your learners and participants.

Responsibility does not transfer to the tool: **you remain accountable**.

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CRITICAL CONSIDERATIONS · SECURITY AND PRIVACY

Security and privacy are decisions you make **before** you build

DO

- ✓ Test with invented or anonymised data.
- ✓ Keep keys and passwords out of the page code.
- ✓ Store data where your approval says.
- ✓ Delete logs you no longer need.

DO NOT

- ✗ Paste real learner or participant data into a chatbot.
- ✗ Share a link that exposes all the data.
- ✗ Believe "it is secure" because the AI says so.
- ✗ Deploy code you have not had checked.

If the data are real, the **stakes** are real.

UNESCO (2023).

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SUMMARY

Three things to take away, and **one thing to do** this week

01

Create, not only use

Chatbots answer. Coding tools build, test and automate.

02

Design for a purpose

Start from the learner activity or the research question.

03

Check before you trust

Accuracy, security, ethics, validity — and stay accountable.

THIS WEEK Pick one repeated task. Write its spec card. Build a tiny version — and check it.

From **using** AI tools to **creating** tools with AI.

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Thank you

Questions and discussion

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